

DATA 3464: Fundamentals of Data Processing

Basic machine learning models

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September 10, 2026

Topic overview

- Some common machine learning tasks and models
- Evaluating model performance
- Limitations and assumptions

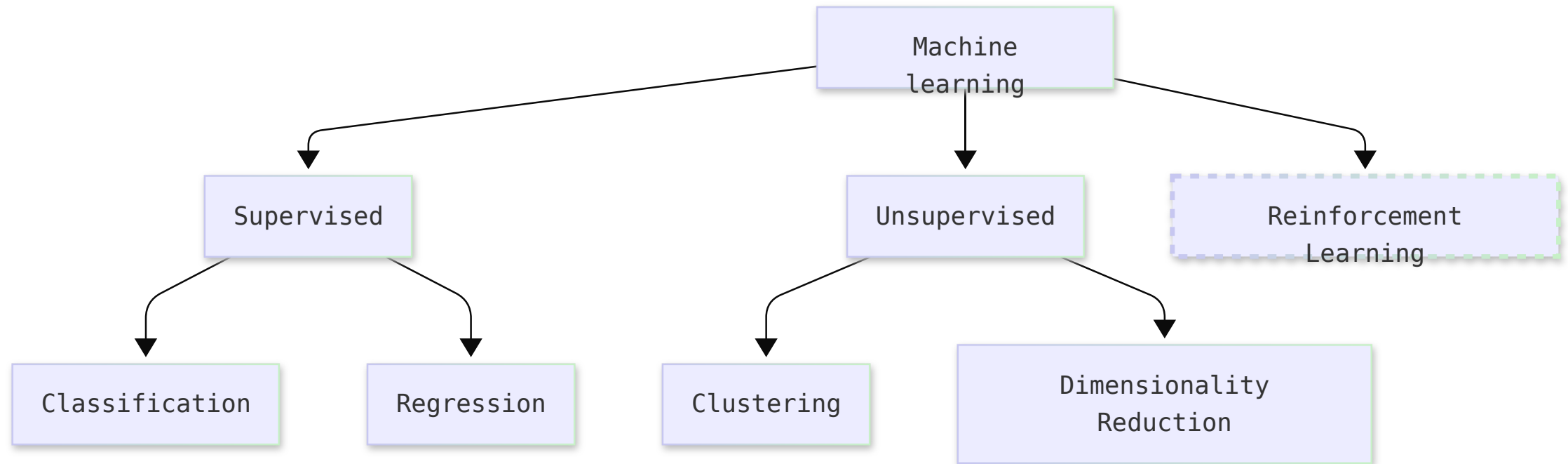
Resources used:

- [Feature Engineering Chapter 1](#)
- Introduction to Machine Learning with Python. Available at [MRU Library](#)
- [Scikit-learn User Guide](#)
- Hands on Machine Learning with Scikit-Learn and Tensorflow/PyTorch. Available at [MRU Library](#)

Machine learning

- To appropriately process the data, we need to know *why* we are doing it and what assumptions we're making
- Modern machine learning toolkits (such as [scikit-learn](#)) are so easy to use, they're easy to use [inappropriately](#)
- Goal: just enough understanding to use basic models **responsibly**

Why are we processing data?



- No reinforcement learning in this course, sorry

A selection of common models

Supervised

- Linear/logistic regression
- Decision trees
- Support vector machines

Unsupervised

- K-means clustering
- Principle component analysis

No free lunch

- A theory-heavy [paper](#) in 1996 showed that there is no one machine learning algorithm that excels in all situations
- Subsequent work has confirmed this, e.g. a [2018 analysis](#)
- Tree-based methods, particularly **gradient boosted trees** *tend* to outperform other algorithms the most, but still have limitations
- What does it mean to "outperform"?

Model evaluation: Classification

- **True positive:** predicted positive, label was positive (TP) ✓
- **True negative:** predicted negative, label was negative (TN) ✓
- **False positive:** predicted positive, label was negative (FP) ✗ (type I)
- **False negative:** predicted negative, label was positive (FN) ✗ (type II)
- **Accuracy** is the fraction of correct predictions, given as:

$$\text{accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision and recall

- **Precision:** Out of all the positive **predictions**, how many were correct?

$$\text{precision} = \frac{TP}{TP + FP}$$

- **Recall:** Out of all the positive **labels**, how many were correct?

$$\text{recall} = \frac{TP}{TP + FN}$$

- **Specificity:** Out of all the negative **labels**, how many were correct?

$$\text{specificity} = \frac{TN}{TN + FP}$$

Confusion matrix

	Predicted Positive	Predicted Negative
Positive Label	TP	FN
Negative Label	FP	TN

- The axes might be reversed, but a good predictor will have strong diagonals
- There's also the **F1 score**, or harmonic mean of precision and recall:

$$F1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

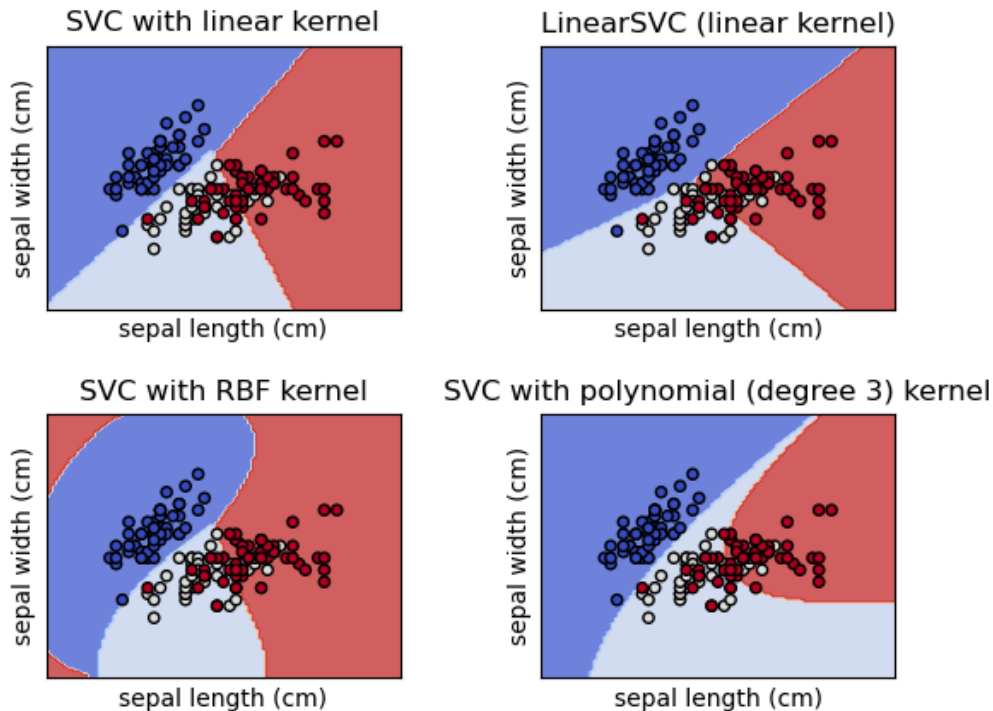
- Better for unbalanced datasets, but still does not capture specificity

ROC Curves

- The **receiver operating characteristic** curve is a plot of the **true positive rate** (recall or sensitivity) vs. **false positive rate** (1 - specificity) as the detection threshold changes
- The diagonal is the same as random guessing
- A perfect classifier would hug the top left corner

Fun fact: the name comes from WWII radar operators, where true positives were airplanes and false positives were noise

Example 1: Support Vector Classifier



- Linear model that finds vector(s) to best separate classes
- "Kernel trick" allows for nonlinear boundaries
- Check out the [SVM Appendix](#) of *Hands-on Machine Learning* by Aurélien Geron for more info

Example 2: Decision Trees

- Family of models including:
 - decision trees
 - random forests
 - gradient boosted decision trees
- Finds thresholds for features that best separates classes
- Controllable through depth parameter

Model evaluation: regression

For a predicted $\hat{\mathbf{y}}$ and actual $\mathbf{y}$, metrics include:

- Mean squared error: $MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$
- Root mean squared error: $RMSE = \sqrt{MSE}$
- Mean absolute error: $MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$
- Mean absolute percentage error: $MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$
- Coefficient of determination: $R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$
(R^2 has some [caveats](#))

Regression: ordinary least squares

- In 1D, estimate modeled as: $\hat{y}_i = wx_i + b$, where i denotes a sample
- Vector form: $\hat{\mathbf{y}} = w\mathbf{x} + b$
- Goal is to minimize the **Mean Square Error**:

$$MSE = \frac{1}{m} \sum_{i=1}^m (\hat{y} - y_i)^2 = \frac{1}{m} \sum_{i=1}^m (b + wx_i - y_i)^2$$

$$MSE = \frac{1}{m} (\hat{\mathbf{y}} - \mathbf{y})^T (\hat{\mathbf{y}} - \mathbf{y})$$

Regression: N-dimensional

- Most of the time we have $N > 1$ **features**
- N-D: $\hat{y}_i = w_1x_{1i} + w_2x_{2i} + \dots + w_nx_{ni} + b$
- Common to use a **design matrix** $\mathbf{X}$ to represent the feature values:

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1n} \\ 1 & x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots & \\ 1 & x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix}$$

where each row is an instance (sample) and each column is a feature.

Regression: N-dimensional

- We can rewrite the estimate in matrix notation:

$$\hat{\mathbf{y}} = \mathbf{X}\theta$$

- The MSE can be written as:

$$MSE = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)^2 = \frac{1}{m} (\mathbf{X}\theta - \mathbf{y})^T (\mathbf{X}\theta - \mathbf{y})$$

- This has a closed form solution, but it is computationally expensive

Regression: Decision trees

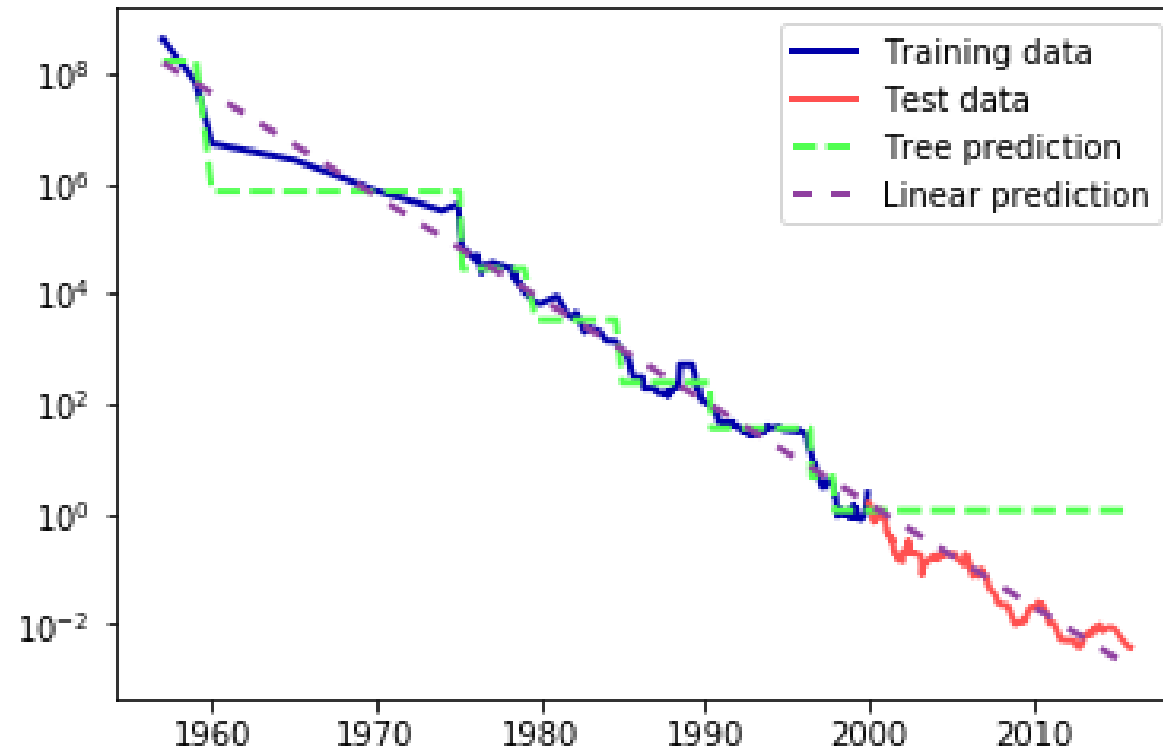
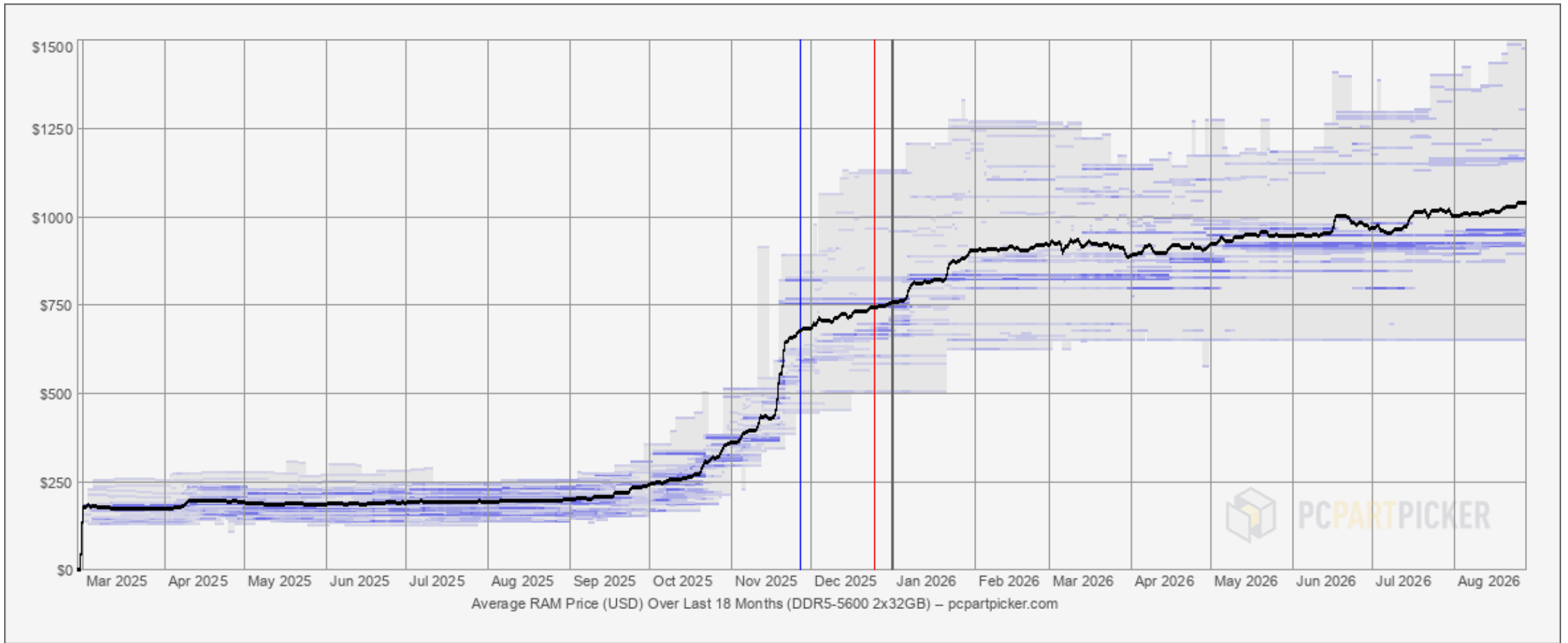


Figure 2-32. Comparison of predictions made by a linear model and predictions made by a regression tree on the RAM price data. Source: Introduction to Machine Learning with Python



Price of DDR5-5600 2X32GB RAM since March 2025 [[PCPartPicker](#)]

Comparison of supervised model families

Linear models

- + Very fast, particularly inference
- + Scalable to large number of features
- + Can model nonlinearity with kernel trick
- + Easy to regularize
- Difficult to interpret
- Sensitive to parameters and preprocessing
- Data needs to be on same scale
- Slow to train on large datasets

Decision Trees

- + Highly explainable
- + Fast to train
- + Few parameters to tune
- + Little preprocessing needed
- + Provides feature importance
- Prone to overfitting
- High variance
- Poor extrapolation

Unsupervised learning

- Learning patterns without labels, such as:
 - Anomaly detection
 - Density estimation
 - Dimensionality reduction
 - **Clustering**

Why is unsupervised learning important?

Clustering: k -means

- Goal: Partition N samples into k clusters
- Naive algorithm:
 - i. Initialize k centroid locations
 - ii. Assign each sample to the nearest centroid
 - iii. Recalculate centroids for each cluster
 - iv. Repeat steps 2-3 until clusters stabilize

$$\sum_{i=0}^N \min_{\mu_j \in C} (\|x_i - \mu_j\|^2)$$

k-means pros and cons

- + Intuitive
- + Easy to implement
- + Relatively scalable ($O(N)$)
- Need to know how many clusters
- Sensitive to initialization
- Not guaranteed to converge to global minimum
- Doesn't like anisotropy and unequal variance

Scikit-learn has a great [reference table](#) of other clustering methods

Categorical data preview

- Samples can take on one of several discrete values or groups
 - **Nominal**: no particular order to the groups
 - **Ordinal**: groups relate to each other in a specific order
- Categories can be represented as strings *or* numeric types
 - Domain knowledge is necessary!

Nominal categories: one-hot encoding

- Categories have no natural relationship
- Create k new features from k categories, very sparse matrix

Animal		cat	dog	rabbit
cat	→	1	0	0
dog	→	0	1	0
rabbit	→	0	0	1

Coming up next

- Basic categorical data encoding
 - Exploring and understanding your data
- Splitting your data
 - Repeatability considerations
 - Stratified sampling
- Assignment 1: Simple modelling